p_cˆsite(Zˆ3) < = 0.4427: summary for a reader

Claim. The critical probability of site percolation on the simple cubic lattice satisfies p_cˆsite(Zˆ3) < = 4427/10000. The best published rigorous upper bound we found is Campanino and Russo's p_c < 1/2 (Ann. Probab. 13 (1985)), listed as 0.5000 in Table 1 of Gomes, Pereira and Sanchis (J. Appl. Probab., 2026).

Argument (pages 2 onward). Tile Zˆ3 by 3x3x3 blocks; in each block take the largest open cluster if it has at least 6 sites (else nothing). A block growth process in the style of van den Berg and Ermakov (1996) explores blocks whose designated clusters touch across shared faces. If, for every history, the children's success indicators stochastically dominate i.i.d. Bernoulli(q), the explored blocks contain a copy of site percolation at q on the coarse lattice, so theta(q) > 0 implies theta(p) > 0. The tree-information lemma reduces "every history" to a worst case over the parent's footprint on the shared face (511 nonempty footprints, 6 entry faces), and Strassen's criterion turns domination into 7579 up-set inequalities per law: a finite check over all 2ˆ27 block states.

Computation. Exhaustive enumeration with exact integer counts; outward-rounded interval sums; exact integer comparison. A chain of seven certificates, each using the previous bound as q, descends from theta(0.501) > 0 (Campanino-Russo's p_c < = 1/2) to p = 0.4427:

link	1	2	3	4	5	6	7
p	0.4634	0.45	0.44525	0.4436	0.443	0.4428	0.4427
q	0.501	0.4634	0.45	0.44525	0.4436	0.443	0.4428

Worst margins are about 1-2 x 10ˆ-4. A variant chain starting at q = 0.68, justified by Wierman's p_cˆsite(Zˆ2) < = 0.679492 (1995) and Zˆ2 contained in Zˆ3, also reaches 0.4427, so Campanino-Russo is not needed.

Checking. The attached package reruns everything: `sh verify-it-yourself/run_checks.sh` (exact fast checks, under a second) and `FULL=1 ...` (the whole exhaustive certificate, about 12 minutes; Go and Python with numpy). Three independent AI reviewers checked the argument, the literature and the computation (one wrote its own exhaustive checker, which agreed on every link); no human has checked it yet, which is why we are writing.

p_cˆsite(Zˆ3) < = 0.4427: a block growth-process coupling with exact certificates

2026-10-18. The coupling lemma and the tree-information lemma follow J. van den Berg and A. Ermakov, "A new lower bound for the critical probability of site percolation on the square lattice", Random Structures Algorithms 8 (1996) 199-212 (growth process and Strassen coupling), written out here for blocks of Zˆ3. Strassen's theorem: a law P on a finite poset stochastically dominates Q (a coupling with X > = Y exists) iff P(A) > = Q(A) for every up-set A (V. Strassen, Ann. Math. Statist. 36 (1965) 423-439).

0. Setting

Site percolation on Zˆ3 with parameter p; P_p, theta(p) = P_p(|C(0)| = infinity). Block B = {0,1,2}ˆ3 (27 sites); B_v = B + 3v for v in Zˆ3. The blocks are disjoint and cover Zˆ3; coarse neighbours v ∼ w iff |v - w|_1 = 1, and then B_v, B_w share a 3 x 3 face: x in B_v and y in B_w are Zˆ3-adjacent iff x lies on that face of B_v and y = x + (w - v) (each face site has exactly one partner, its mirror image across the face). Sites of blocks that are not coarse neighbours are never Zˆ3-adjacent.

eps(v) in {0,1}ˆB is the configuration of B_v translated to B. G(eps) is the largest open component of eps (connectivity inside B); ties are broken by the larger sum over the six faces of the number of the component's sites on that face, then by the first component found in site order; and G = empty set if that component has fewer than s = 6 sites (or eps has no open site). Any deterministic rule giving a connected set or the empty set would do.

For coarse neighbours v, w: eta(v, w) = 1 iff some site of G(eps(v)) + 3v is Zˆ3-adjacent to some site of G(eps(w)) + 3w. Writing T_F(eps) for the trace of G(eps) on face F, eta(v, w) = 1 iff T_F(eps(v)) and the mirror of T_{F'}(eps(w)) meet, where F, F' are the faces of B_v, B_w on their common boundary. eta is a function of (eps(v), eps(w)).

1. The growth process

R is the event that eps(0) is all open (P_p(R) = pˆ27 > 0). Under P_p(. | R): xi(0) = 1. Fix a total order on Zˆ3 with least element 0. At each step select the least v with xi(v) = 1 that has a coarse neighbour without a xi-value; those neighbours are v's children; each child u gets xi(u) = eta(v, u).

Claim. If xi(u) = 1 then every site of G(eps(u)) + 3u is in the open cluster of 0. (Induction along the parent chain: G(eps(u)) is connected and contains a site adjacent to a site of G(eps(parent)), which is in the cluster; B_0 is all open and connected and contains 0.)

2. Coupling lemma (Lemma 1 of the square-site proof, for blocks)

Lemma 1. Suppose, under P_p(. | R), at every step and for every history, the conditional law of the children's xi-values stochastically dominates Bernoulli(q)ˆchildren. Then theta_{Zˆ3}(q) < = P_p(survival | R) < = theta(p)/pˆ27.

Proof. Build zeta alongside xi, under P_p(. | R). At the root, zeta(0) ∼ Bernoulli(q) independently, so zeta(0) < = xi(0) = 1. At each step, given the history of both processes, use Strassen's theorem to choose zeta on the children with law Bernoulli(q)ˆchildren, independent of the history, and zeta < = xi on the children; coarse vertices never assigned a xi-value get fresh independent Bernoulli(q) values.

zeta is i.i.d. Bernoulli(q). The zeta values on a step's children are drawn from a Strassen coupling Q_h(. | xi on the children), using fresh auxiliary uniforms, where Q_h may depend on the history h. Earlier zeta values are functions of earlier xi-values and auxiliary uniforms, which are independent of the block states. So the law of the new xi-values given (h, earlier zeta) is their law given h, and the new batch of zeta is exactly Bernoulli(q)ˆchildren given everything before it; the number of children is a function of h.

For the whole field, pre-sample independent Bernoulli(q) coins Y(v), and after n steps form the hybrid field: zeta on the vertices assigned so far, Y elsewhere. Each hybrid field is i.i.d. Bernoulli(q) by induction on n, and the hybrid fields converge pointwise to zeta, so every finite-dimensional law of zeta is the product law.

If the process dies out (finitely many steps), no vertex with xi = 1 has a neighbour without a xi-value, so every coarse neighbour u of a xi = 1 vertex received its xi-value at some step, with zeta(u) <= xi(u). By induction along any zeta-open path from 0 (zeta(0) <= xi(0)), every vertex of the path has xi = 1, so the zeta-open cluster of 0 lies in the finite set {xi = 1}. If the process survives, it has infinitely many steps, each selecting a distinct block with xi = 1; by the Claim, the open cluster of 0 meets all of them, so it is infinite. Hence theta_{Z^3}(q) <= P_p(survival | R) <= P_p(|C(0)| = infinity | R) <= theta(p)/P_p(R). []

(The die-out/survival split is needed: for a general order, a vertex with xi = 1 need not ever be selected when the process survives.)

3. Tree information (Lemma 2 of the square-site proof)

The history is R together with the event that eta(e) = x_e on the edges of the exploration tree. Children are blocks without a xi-value, so their eps are independent of the history. Hence the children's xi law given the history is sum_beta w(beta) nu_beta, where w is the posterior law of eps(b2) (b2 the selected block, c its parent) and nu_beta = prod over children i of Bernoulli(r_i(T_i(beta))), with r_i(T) = P_p(T meets the mirror of the trace of G(eps') on the facing face) for independent eps'. By message passing on the tree, w(beta) ∝ pi(beta) sum_gamma 1[eta(gamma, beta) = 1] mu_c(gamma), mu_c a probability law on the states of c (the point mass at the all-open state when c = 0). Grouping gamma by U = mirror of T_F(gamma): w = sum_U lambda_U w_U, with w_U(beta) ∝ pi(beta) 1[T_in(beta) meets U], over nonempty U in the entry face.

4. Reduction to finitely many checks

The set of laws on {0,1}^k that dominate Bernoulli(q)^k is convex, and domination survives projection to a subset of coordinates. So Lemma 1's hypothesis holds if: (i) at the root step (eps(0) all open, U = the full face for each child): r_f(full face) >= q for each face f; (ii) for each entry face F_in and each nonempty U contained in F_in, the law of the 5 children's eta under w_U dominates Bernoulli(q)^5. By Strassen's theorem this is: for every up-set A of {0,1}^5, law_U(A) >= q-measure of A. Both are finite computations over the 2^27 block states, done in interval arithmetic with outward rounding.

5. The chain

Each link j is a run of the checks of section 4 at a rational pair (p_j, q_j); if they pass, Lemma 1 gives theta(p_j) >= p_j^27 theta(q_j), so theta(q_j) > 0 implies theta(p_j) > 0 and p_c <= p_j.

Base: theta(q_1) > 0 for q_1 = 501/1000, because p_c^site(Z^3) <= 1/2 (Campanino and Russo 1985, the comparison with the triangular lattice; their strict inequality is not needed). Links (all checks passed in exact arithmetic; outputs/chainB_summary.txt and outputs/chainA_summary.txt):

link	p_j	q_j	worst up-set margin
1	2317/5000 = 0.4634	501/1000	1.94e-4
2	9/20 = 0.45	2317/5000	1.12e-4
3	1781/4000 = 0.44525	9/20	1.14e-4
4	1109/2500 = 0.4436	1781/4000	2.04e-4
5	443/1000 = 0.443	1109/2500	1.66e-4
6	1107/2500 = 0.4428	443/1000	1.99e-4

7	4427/10000 = 0.4427	1107/2500	1.23e-4

(An alternative start, q_1 = 1/2 with p_1 = 463/1000 and p_2 = 4499/10000, also passes; it needs Campanino-Russo's strict inequality.) Every link also checks the root condition r_f(full face) >= q_j exactly (outputs/*_log.txt, "root:" lines).

Theorem. p_cˆsite(Zˆ3) <= 4427/10000 = 0.4427.

Errata (cosmetic, found by the panel; the object is unchanged from what the panel checked)

1. object/code/cert/go.mod says `go 1.26.4` (the toolchain used), while the documents say Go >= 1.21. The source needs nothing newer than Go 1.21. With an older Go, either let Go fetch the toolchain (GOTOOLCHAIN=auto, the default) or change that line to `go 1.21` before building.
2. object/code/chain.py always ends with "given Campanino-Russo's p_c < 1/2 for link 1", even when a run starts at q1 = 501/1000 (which needs only p_c <= 1/2) or at another q. The certificates do not depend on that text; PROOF.md section 5 states the base correctly.
3. object/code/chain.py runs code/cert/cert.exe and the fixed arguments 3 3 3 6 6 (block 3x3x3, 6 workers, minimum size 6). On Linux or macOS build with `go build -o cert.exe .` (the name is arbitrary).
4. PROOF.md calls the finite checks "interval arithmetic" in section 4 and "exact arithmetic" in section 5: the laws are outward-rounded intervals and the final comparison is exact; both statements refer to that one procedure.
5. how-we-built-it/outputs/chainA_summary.txt: a line "[exited with code 1]" in the panel's copy came from the shell capture (tee could not open its log file at start), not from chain.py; every step exits 0 (chainA_log.txt). The copy here has that line removed.
6. "Site order" in the tie-break means increasing site index id = 9x + 3y + z (object/code/cert/main.go). Any deterministic rule giving a connected set or the empty set makes the proof work.